

Week 4: Dense Retrieval Baseline (sampled corpus)

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1. Objective

Build a Sentence-Transformers + FAISS dense retrieval baseline on a **sampled** sub-corpus of MS MARCO Passage, and compare it head-to-head against BM25 rebuilt on the **same** sample. We deliberately skip encoding the full 8.8M corpus on a CPU laptop; the trade-off is spelled out below.

2. Setup

- **Encoder:** sentence-transformers/all-MiniLM-L6-v2 (no fine-tuning, generic semantic similarity).
- **Sample size:** 50,000 passages, qrels-anchored.
- **Index:** faiss.IndexFlatIP over L2-normalised embeddings — exact inner-product search $\equiv$ cosine similarity. No ANN approximation, so the numbers reflect the encoder, not the index.
- **Comparison BM25:** rebuilt on the **same** 50,000-passage sample with bm25s ($k_1=1.50$, $b=0.75$). This is the apples-to-apples comparison; the W2 full-corpus BM25 baseline ($\text{MRR}@10 = 0.1703$) is *not* directly comparable to either column below.
- **Eval queries:** dev/small (6980 of 6980 queries have ≥ 1 relevant doc landing in the sample).

3. Sampling note (important caveat)

A uniform random 50k sample from 8.8M passages would put almost no relevant doc in the pool, so both retrievers would score near zero and the comparison would be meaningless. We use **qrels-anchored sampling** instead:

- Include every dev/small relevant doc_id (7437 unique docs).
- Fill the rest with random distractors from the full corpus's doc_id pool (seed = 42).

The relevant doc is therefore *always* present in the pool, which makes absolute MRR / Recall / nDCG higher than any "real" retrieval setting. **The valid comparison is BM25-on-sample vs dense-on-sample**, not either column vs the W2 full-corpus number.

4. Results

Metric	BM25 (sample)	Dense (sample)	Δ (dense – BM25)
MRR@10	0.6948	0.8830	+0.1882
nDCG@10	0.7270	0.9041	+0.1771
Recall@100	0.9338	0.9946	+0.0608
Recall@1000	0.9686	0.9991	+0.0305

Wall-clock:

- Encode corpus: 810.5 s (16.21 ms / passage)
- Dense search (6980 queries $\times$ top-1000): 17.6 s
- BM25-sample build: 4.5 s
- BM25-sample search: 8.8 s

5. Qualitative examples

909176 — what treatments are available for addicted individuals

- Dense first-relevant rank: **1**
- BM25 first-relevant rank: **>10**
- Relevant doc id(s): 7939267

1081086 — what food is good for low carb diet

- Dense first-relevant rank: **1**
- BM25 first-relevant rank: **>10**
- Relevant doc id(s): 7148498

632455 — what does arthralgia mean in medical terms

- Dense first-relevant rank: **1**
- BM25 first-relevant rank: **>10**
- Relevant doc id(s): 7587963

46095 — average time for hair to grow

- Dense first-relevant rank: **1**
- BM25 first-relevant rank: **>10**
- Relevant doc id(s): 1790005

647503 — what does sbp mean

- Dense first-relevant rank: **1**
- BM25 first-relevant rank: **>10**
- Relevant doc id(s): 7632291

6. Discussion

Concrete observations from this run (sampled, generic encoder, no fine-tune):

- **MRR@10**: dense wins by 0.1882 (dense 0.8830 vs BM25 0.6948).
- **nDCG@10**: dense wins by 0.1771 (dense 0.9041 vs BM25 0.7270).
- **Recall@100**: dense wins by 0.0608 (dense 0.9946 vs BM25 0.9338).
- **Recall@1000**: dense wins by 0.0305 (dense 0.9991 vs BM25 0.9686).
- Caveat: every dev relevant doc is in the sampled pool by construction, so absolute numbers are upper-bounded and will drop on a larger / non-anchored sample.

7. Limitations

- **Sampled corpus**: every dev/small relevant doc is unconditionally included. Numbers here are systematically higher than they would be on the full 8.8M corpus.
- **No fine-tuning**: sentence-transformers/all-MiniLM-L6-v2 is pretrained for generic semantic similarity, not MS MARCO retrieval specifically. A fine-tuned encoder (e.g. msmarco-MiniLM-L6-cos-v5) would close most of the gap with leaderboard dense baselines.
- **No ANN approximation**: IndexFlatIP is exact and slow on big corpora. For the full corpus we'd want IVF-PQ / HNSW.
- **Single seed**: distractor sample is deterministic but not averaged.

8. Next

- Hybrid retrieval (BM25 + dense score fusion, e.g. RRF or weighted sum).
- Cross-encoder reranking over the BM25 top-100 (ms-marco-MiniLM-L-6-v2).
- Switch the encoder to a MS MARCO-tuned variant and re-run on the same sample.
- Scale the sample to 200k–500k passages to track how the BM25 ☐ dense gap shifts as the pool grows (relevant doc density drops).